

Loan Default Risk Analysis

Student portfolio project

Explore income, credit, loan amounts, and default rates for risk-focused decision support.

project_id: loan_default_risk | template_id: root_cause | source_rows: 48 | prepared_rows: 48 | generated_at: 2026-08-11T06:45:05.140163+00:00

Executive summary

Loan Default Risk Analysis analyzed 48 rows across 7 fields. The reusable root_cause BI template presents source-backed KPIs, drivers, and detail tables. Total loan amount is \$4.1M with a transparent synthetic validation fixture or uploaded source.

Headline metrics

Rows analyzed

48

Total loan amount

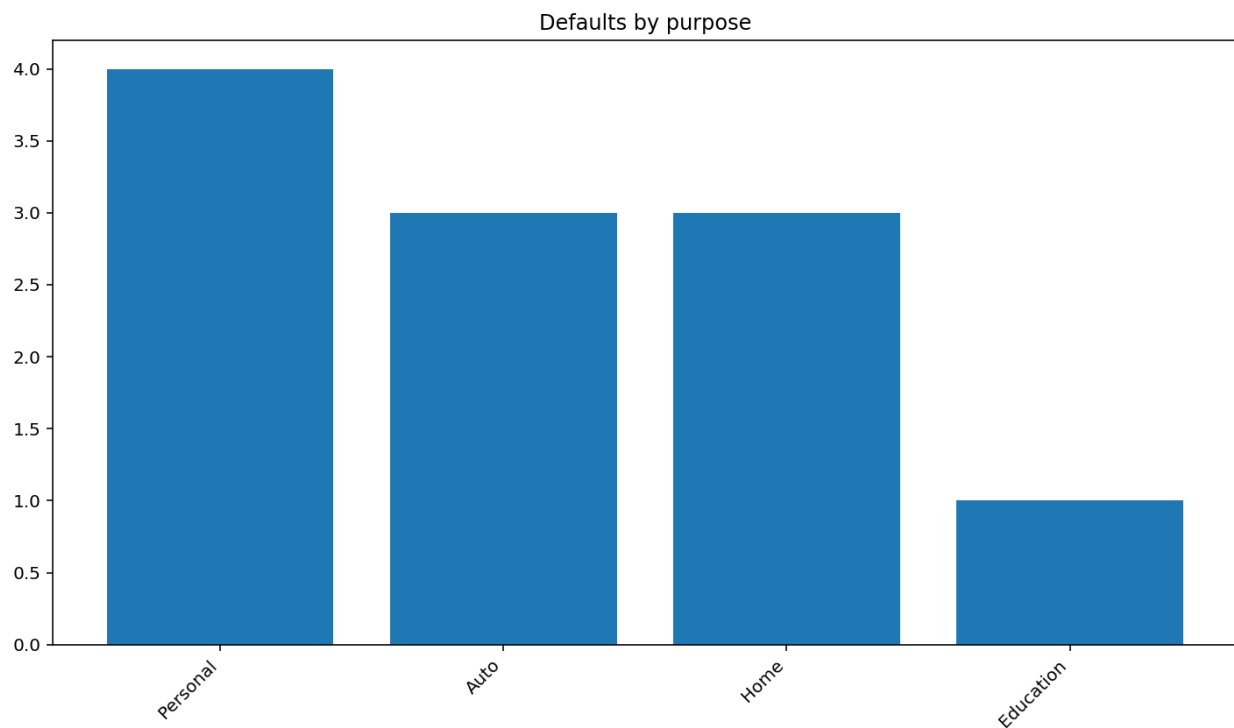
\$4.1M

Average loan amount

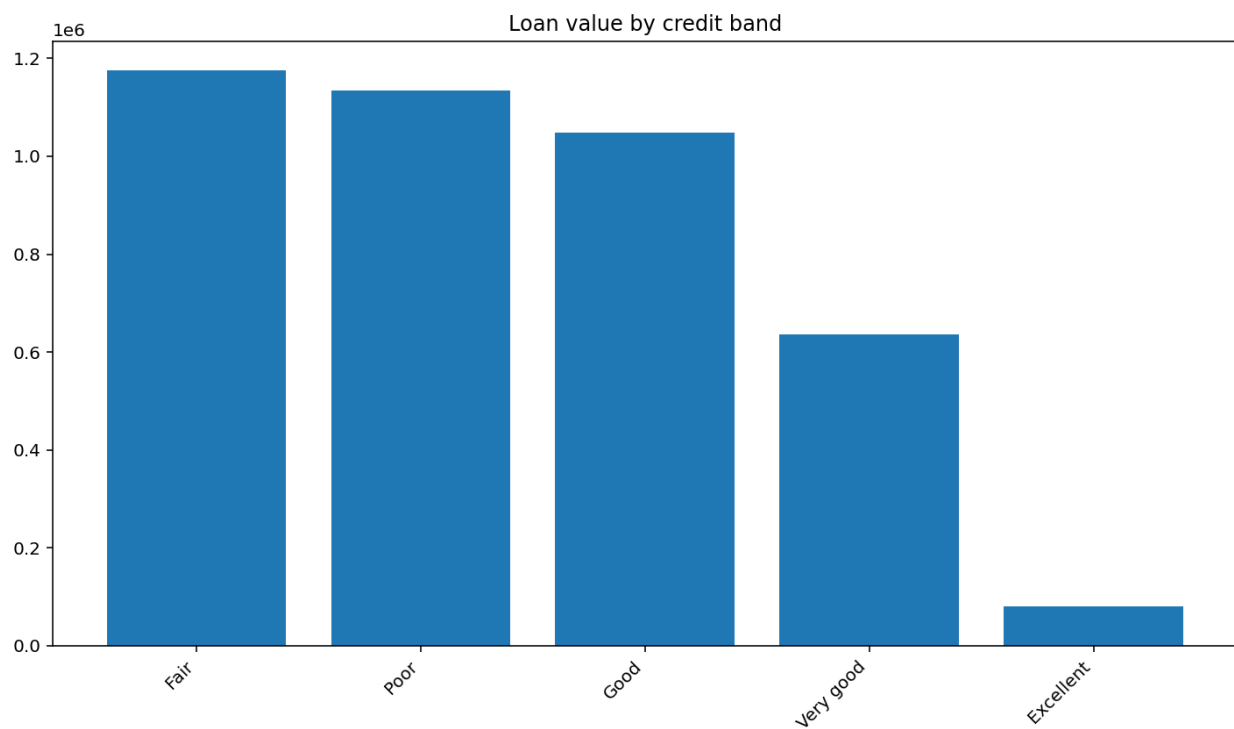
\$84.9k

Drivers and trends

Defaults by purpose



Loan value by credit band



Detail by purpose

purpose	count	sum	mean
Personal	12	4.0	0.3333333333333333
Auto	12	3.0	0.25

purpose	count	sum	mean
Home	12	3.0	0.25
Education	12	1.0	0.08333333333333333

Detail by credit_band

credit_band	count	sum	mean
Fair	14	1176241.0	84017.21428571429
Poor	12	1134444.0	94537.0
Good	11	1048074.0	95279.45454545454
Very good	8	635864.0	79483.0
Excellent	3	80641.0	26880.333333333332

Key findings

credit_band explains substantial variation in credit_score ($\eta^2=0.965$).; income is nearly unique per row (100.0%).; loan_amount is nearly unique per row (100.0%).; loan_id is nearly unique per row (100.0%).; defaulted has 22.9% IQR outliers.; credit_band explains substantial variation in defaulted ($\eta^2=0.182$).; credit_band explains substantial variation in income ($\eta^2=0.150$).; defaulted is strongly right skewed (skew=1.33).

Caveats and assumptions

Fixtures are synthetic validation data. Uploaded datasets remain the source of truth; field aliases are recorded in the build metadata.